

EBN111 EKSAMEN ANTWOORDBLAD/ EXAM ANSWER SHEET**8 JUNE 2009**

Student se besonderhede/Student's details	
Van en voorletters: Surname and initials:	Studentenommer: Student number:
Graadkursus: Degree Course:	
Tel. nr.: Tel no.:	

MEMORANDUM

MARKS									
A1	/30	B1	/30	B2	/12	B3	/18	Final	/90

ANSWERS: AFDELING A / SECTION A

VRAAG 1 – QUESTION 1											
1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10	1.11	1.12
B	B	A	A	D	D	E	B	D	D ₍₂₎	D ₍₂₎	C ₍₂₎
1.13	1.14	1.15	1.16	1.17	1.18	1.19	1.20	1.21	1.22	1.23	1.24
C	B	F	E	D	A	A	C	B	A ₍₂₎	D ₍₂₎	C ₍₂₎

ANSWERS: AFDELING B / SECTION B

VRAAG 1 – QUESTION 1								Marks	
1.1	$i_1 = -3A$		$i_2 = -1.5A$		$v_1 = 9V$		$v_2 = 35V$	$p = -105W$	5
1.2	$a) Req = 20\Omega$				$b) Req = 30\Omega$		$c) Req = 22\Omega$		3
1.3	$R_1 = 5\Omega$			$R_2 = 5\Omega$		$R_3 = 2.5\Omega$		$R_{xy} = 50\Omega$	4
1.4	$a_1 = 5$	$a_2 = 6$	$a_3 = 36$	$b_1 = -1$	$b_2 = 1$	$b_3 = -10$	$v_o = 8.73$		4
1.5	$v_o = 50V$								3
1.6	$I_s = 448.5\mu A$				$V_1 = 29.76V$				4
1.7	$R_{TH} = 1.5\Omega$		$V_{ab}(12V \text{ source}) = 2V$			$V_{ab}(3A \text{ source}) = -3V$			4
1.8	$R_{TH} = 7.06\Omega$								3

VRAAG 2 – QUESTION 2

2.1	$V_0 = -7.5 \text{ V} \quad (2)$	$V_a = 0.2 \text{ V} \quad (2)$
2.2	$V_1 = 1.8 \text{ V} \quad (1)$	$V_0 = 7.2 \text{ V} \quad (2)$
2.3	$i(0) = -0.36 \text{ A} \quad (2)$	$W_c = 23.9 \mu\text{J} \quad (1)$
2.4	$p = -0.757 \text{ W} \quad (2)$	

VRAAG 3 – QUESTION 3

3.1	(a) $v(t) = 43.6 \cos(\omega t - 6.6^\circ) \text{ V} \quad (2)$		
	(b) $\bar{V} = 8 \angle -20^\circ \text{ V} \quad (1)$		
	(c) $\bar{I} = 60 \angle -110^\circ \text{ mA} \quad (1)$		
	(d) $\bar{V}_1 \quad (1)$		
	(e) Determinant = $-6 - j11 \quad (2)$		
3.2	$\bar{a} = 6 + j3$		$\bar{b} = -j5 \quad (2)$
	$\bar{c} = -j5$	$\bar{d} = 5 + j5$	$\bar{e} = 5 \angle 0^\circ = 5 \quad (2)$
3.3	$i(t) = 8.94 \cos(\omega t - 33.43^\circ) \text{ A} \quad (3)$		
3.4	$\bar{I}_N = 3.39 \angle 32^\circ \text{ A} \quad (2)$		$\bar{Z}_N = 10 \angle 26^\circ \Omega \quad (2)$